

THE NATURE & PROPAGATION OF LIGHT

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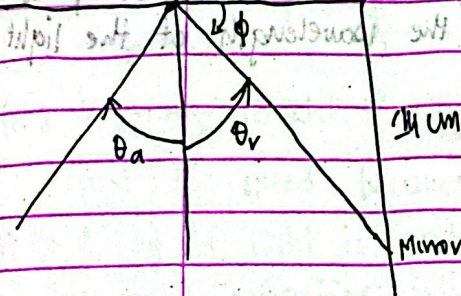
Exercises

77.1 Two plane mirrors intersect at right angles. A laser beam strikes the first of them at a point 11.5 cm from their point of intersection, as shown in Fig. E77.1. For what angle of incidence at the first mirror will this ray strike the midpoint of the second mirror (which is 28.0 cm long) after reflecting from the first mirror? FIG. E77.1

Sol'n: Reflection, $\theta_r = \theta_a$

$$\phi = 50.6^\circ$$

Mirror 11.5 cm



$$\tan \phi = \frac{14}{11.5} = 50.6^\circ$$

$$\theta_r = 90^\circ - \phi$$

$$\theta_r = 39.4^\circ$$

$$\theta_r = \theta_a = 39.4^\circ$$

77.2 The vitreous humor, a transparent, gelatinous fluid that fills most of the eyeball, has an index of refraction of 1.34. Visible light ranges in wavelength from 380 nm (violet) to 750 nm (red) as measured in air. This light travels through the vitreous humor and strikes the rod and cones at the surface of the retina. What are the ranges of (a) the wavelength, (b) the frequency, and (c) the speed of the light just as it approaches the retina within the vitreous humor?

Sol'n: $n = \frac{c}{v}$, $v = f\lambda$, $\lambda = \frac{\lambda_0}{n}$

a) $\lambda_r = \frac{\lambda_0}{n} = \frac{400 \text{ nm}}{1.34} = 299 \text{ nm}$ $\lambda_v = \frac{\lambda_0}{n} = \frac{700 \text{ nm}}{1.34} = 522 \text{ nm}$

the range is 299 nm to 522 nm

b) calculate the frequency in air, where $v = c = 3.00 \times 10^8 \text{ m/s}$

$f_r = \frac{c}{\lambda_r} = \frac{3.00 \times 10^8 \text{ m/s}}{700 \times 10^{-9} \text{ m}} = 4.29 \times 10^{14} \text{ Hz}$

$f_v = \frac{c}{\lambda_v} = \frac{3.00 \times 10^8 \text{ m/s}}{400 \times 10^{-9} \text{ m}} = 7.50 \times 10^{14} \text{ Hz}$

c) $v = \frac{c}{n} = \frac{3.00 \times 10^8 \text{ m/s}}{1.34} = 2.24 \times 10^8 \text{ m/s}$

77.3) A beam of light has a wavelength of 650 nm in vacuum, (a) What is the speed of this light in a liquid whose index of refraction at this wavelength is 1.47? (b) What is the wavelength of these waves in the liquid?

sol'n: a.) $n = \frac{c}{v} \Rightarrow v = \frac{c}{n} = \frac{2.998 \times 10^8 \text{ m/s}}{1.47} = 2.04 \times 10^8 \text{ m/s}$

b.) $\lambda = \frac{\lambda_0}{n} = \frac{650 \text{ nm}}{1.47} = 442 \text{ nm}$

77.4 light with a frequency of $5.80 \times 10^{14} \text{ Hz}$ travels in a block of glass that has an index of refraction of 1.52. What is the wavelength of the light (a) in vacuum and (b) in glass?

sol'n: $c = f\lambda_0 \Rightarrow \lambda_0 = \frac{c}{f}$
a.) $\lambda_0 = \frac{c}{f} = \frac{3.00 \times 10^8 \text{ m/s}}{5.80 \times 10^{14} \text{ Hz}} = 517 \text{ nm}$

b.) $\lambda = \frac{\lambda_0}{n} = \frac{517 \text{ nm}}{1.52} = 340 \text{ nm}$

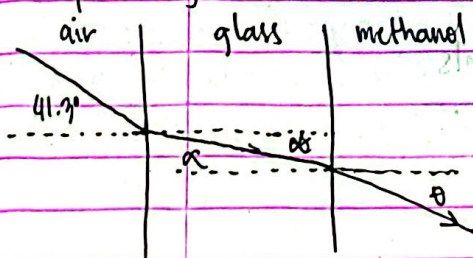
77.5. A light beam travels at $1.94 \times 10^8 \text{ m/s}$ in quartz. The wavelength of light in quartz is 355 nm. (a) What is the index of refraction of quartz at its wavelength? (b) If this same light travels through air, what is its wavelength there?

sol'n: $n = \frac{c}{v}$ and $\lambda = \frac{\lambda_0}{n}$
a.) $n = \frac{c}{v} = \frac{3.00 \times 10^8 \text{ m/s}}{1.94 \times 10^8 \text{ m/s}} = 1.55$ b.) $\lambda_0 = n\lambda = (1.55)(355 \text{ nm}) = 550 \text{ nm}$

77.10 (a) A tank containing methanol has walls 2.50 cm thick made of glass of refractive index 1.550. Light from the outside air strikes the glass at 41.7° angle with the normal to the glass. Find the angle the light makes with the normal in the methanol. (b) The tank is emptied and refilled with an unknown liquid. If light incident at the same angle as in part (a) enters the liquid in the tank at an angle of 20.2° from the normal, what is the refractive index of the unknown liquid?

sol'n: Applying Snell's law at both interfaces.

The path ray is sketched in Fig 77.10 below. Gives $n = 1.729$ for the methanol



(a) At the air-glass interface $1.00 \sin 41.7^\circ = n \sin \theta$
At the glass-methanol interface $n_{\text{glass}} \sin \theta = 1.729 \sin \theta$
Combining these two equations gives

$\sin 41.7^\circ = 1.729 \sin \theta$ and $\theta = 29.8^\circ$

(b) The same figure applies. apply for part (a), except $\theta = 20.2^\circ$

$$1.00 \sin 41.7^\circ = n \sin 20.2^\circ \text{ and } n = 1.91$$

99.17 In a material having an index of refraction n , a light ray has frequency f and wavelength λ , and speed v . What are the frequency, wavelength, and speed of this light (a) in vacuum and (b) in a material having refractive index n' ? For each case, express your answers in terms of only f , λ , v , n , and n' .

$$\text{Sol'n: } \lambda = \frac{\lambda_0}{n}, v = \frac{c}{n}$$

$$(a) \text{ frequency: } f = \frac{c}{\lambda_0}, \text{ wavelength } \lambda_0 = n\lambda, \text{ speed } v = \frac{c}{n} \Rightarrow c = nv$$

$$(b) \text{ frequency is still } f = \frac{c}{\lambda_0}, \text{ wavelength becomes } \lambda' = \frac{\lambda_0}{n'} = \frac{n\lambda}{n'} = \left(\frac{n}{n'}\right)\lambda$$

$$\text{and the speed becomes } v' = \frac{c}{n'} = \frac{nv}{n'} = \left(\frac{n}{n'}\right)v$$

99.15 A ray of light is incident on the plane surface separating two sheets of glass with refractive indices 1.70 and 1.58. The angle of incidence is 62.0° , and the ray originates in the glass with $n = 1.70$. Compute the angle.

$$\text{Sol'n: } n_a \sin \theta_a = n_b \sin \theta_b, n_a = 1.70, \theta_a = 62.0^\circ, n_b = 1.58$$

$$\sin \theta_b = \frac{n_a}{n_b} \sin \theta_a = \frac{1.70}{1.58} \sin 62.0^\circ = 0.950, \theta_b = 71.8^\circ$$

99.16 A flat piece of glass covers the top of a vertical cylinder that is completely filled with water. If a ray of light traveling in the glass is incident on the interface with water at an angle of $\theta_a = 36.2^\circ$, the ray refracted into the water makes an angle of 49.8° with the normal to the surface. What is the smallest value of the incident angle θ_a for which none of the rays refracts into the water?

$$\text{Sol'n: } n_a \sin \theta_a = n_b \sin \theta_b$$

$$n_g \sin 36.2 = n_w \sin 49.8 \Rightarrow n_g = (1.33) \left(\frac{\sin 49.8^\circ}{\sin 36.2^\circ} \right) = 1.724$$

$$\text{Now find } \theta_{\text{crit}}, n_g \sin \theta_{\text{crit}} = n_w \sin 90^\circ \Rightarrow \sin \theta_{\text{crit}} = \frac{1.33}{1.724}, \theta_{\text{crit}} = 50.6^\circ$$

99.18 A beam of light is traveling inside a solid glass cube having index of refraction 1.53. It strikes the surface of the cube from the inside. (a) If the cube is in air, at what minimum angle with the normal inside the glass will this light not enter the air at this surface? (b) What would be the minimum angle in part (a) if the cube were immersed in water?

$$\text{Sol'n: } n_{\text{glass}} \sin \theta_{\text{crit}} = n_{\text{air}} \sin 90^\circ$$

$$(a) 1.53 \sin \theta_{\text{crit}} = 1.00 (1) ; \theta_{\text{crit}} = 40.8^\circ$$

(b) $1.57 \sin \theta_{\text{crit}} = 1.57 \sin 90^\circ$; $\theta_{\text{crit}} = 60.6^\circ$

77.19 The critical angle for total internal reflection at a liquid-air interface is 42.5° . (a) If a ray of light traveling in the liquid has an angle of incidence at the interface of 75.0° , what angle does the refracted ray in the air make with the normal? (b) If a ray of light traveling in air has an angle of incidence at the interface of 75.0° , what angle does the refracted ray in the liquid make with the normal?

Sol'n: $\theta_a = \theta_{\text{crit}}$ when $\theta_b = 90^\circ$, so $n_{\text{liq}} \sin \theta_{\text{crit}} = 1.0 \sin 90^\circ$
 $n_{\text{liq}} = \frac{1}{\sin \theta_{\text{crit}}} = \frac{1}{\sin 42.5^\circ} = 1.48$

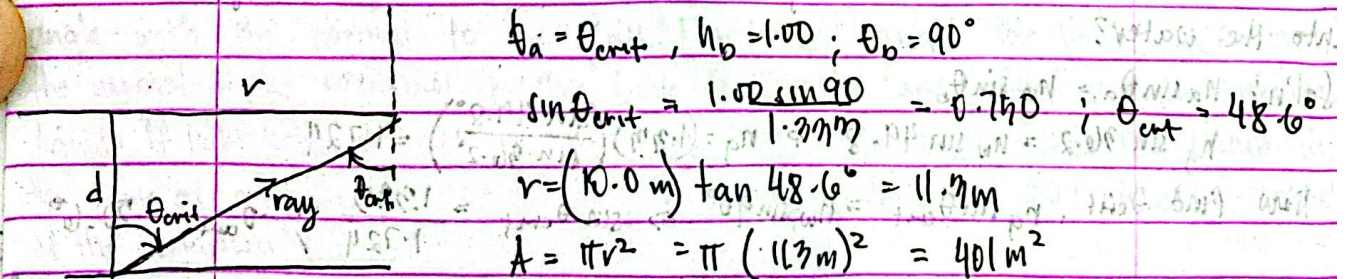
(a) $n_a \sin \theta_a = n_b \sin \theta_b \rightarrow (a = \text{liquid}, b = \text{air})$

$\sin \theta_b = \frac{n_a \sin \theta_a}{n_b} = \frac{1.48 \sin 75^\circ}{1.0} = 0.8489$; $\theta_b = 58.1^\circ$

(b) $\sin \theta_b = \frac{n_a \sin \theta_a}{n_b} = \frac{1.0 \sin 75^\circ}{1.48} = 0.7876$; $\theta_b = 22.8^\circ$

77.20 At the very end of Wagner's series of operas Ring of the Nibelung Brünnhilde takes the golden ring from the finger of the dead Siegfried and throws it into the Rhine, where it sinks to the bottom of the river. Assuming that the ring is small enough compared to the depth of the river to be treated as a point and that the Rhine is 10.0 m deep where the ring goes in, what is the area of the largest circle at the surface over which light from the ring could escape from the water?

Sol'n: The diagram is shown below



77.26 A beam of light strikes a sheet of glass at an angle of 57.0° with the normal in air. You observed the red light makes an angle of 38.1° with the normal in the glass, while violet light makes a 36.7° angle. (a) What are the indexes of refraction of this glass for these colors of light? (b) What are the speeds of red and violet light in the glass?

Sol'n: $n_a \sin \theta_a = n_b \sin \theta_b$; when $n_a = 1.00$; $\theta_a = 57.0^\circ$; $\theta_b = 38.1^\circ$ (red)
 $n_{\text{red}} = \frac{\sin 57.0^\circ}{\sin 38.1^\circ} = 1.47$
 $v = \frac{c}{n}$

(a) $n_b = \frac{n_a \sin \theta_a}{\sin \theta_b} = \frac{1 \sin 57.0^\circ}{\sin 38.1^\circ} = 1.26$

Violet: $n_b = \frac{1.0 \sin 57}{\sin 36.7} = 1.40$

(b) red: $v = \frac{c}{n} = \frac{3.00 \times 10^8 \text{ m/s}}{1.26} = 2.21 \times 10^8 \text{ m/s}$

Violet: $v = \frac{c}{n} = \frac{3.00 \times 10^8 \text{ m/s}}{1.40} = 2.14 \times 10^8 \text{ m/s}$

97.27 Un-polarized light with intensity I_0 is incident on two polarizing filter. The axis of the first filters makes an angle of 40° with the vertical, and the axis of the second filter is horizontal. What is the intensity of the light after it has passed through the second filter?

Sol'n: $I = I_0 \cos^2 \phi$

1st filter: $I = \frac{1}{2} I_0$ 2nd filter: $I = \frac{1}{2} (I_0) \cos^2 \phi$

$I = \frac{1}{2} (I_0) \cos^2 70 = 0.775 I_0$

97.28 (a) At what angle above the horizontal is the sun if sunlight, reflected from the surface of the calm lake is completely polarized? (b) What is the plane of the electric-field vector in the reflected light?

Sol'n: $\theta_a = \theta_p$, $\tan \theta_p = \frac{n_b}{n_a}$, $n_a = 1.00$, $n_b = 1.77$

(a) $\tan \theta_p = \frac{1.77}{1.0} = 1.77$; $\theta_p = 57.1^\circ$. The sunlight is incident at an angle of $90^\circ - 57.1^\circ = 32.9^\circ$ above the horizontal.

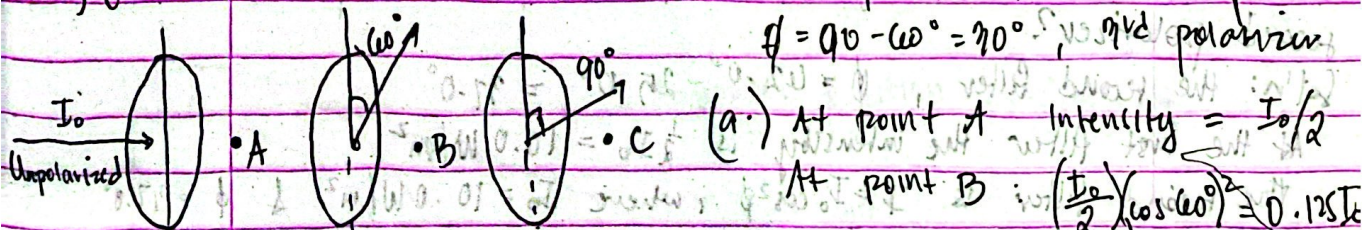
(b) The plane of the electric-field vector in the reflected light is horizontal.

97.29 A beam of intensity light, of I_0 passes through a series of ideal polarizing filters with their polarizing directions turned into various angles as shown in Figure 97.29 (a) What is the light intensity (in terms of I_0)

at points A, B and C? (b) If we remove the second filter/middle filter, what will be the light intensity at point C?

Fig 97.29 Sol'n: $\phi = 60^\circ$, for each polarizer

$\phi = 90^\circ - 60^\circ = 30^\circ$, 3rd polarizer



(a) At point A Intensity = $I_0/2$

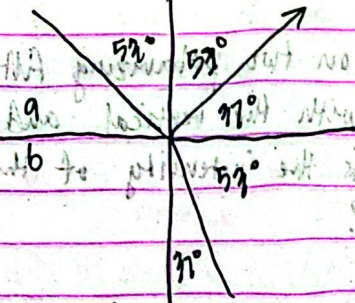
At point B: $(\frac{I_0}{2}) (\cos 60^\circ)^2 = 0.125 I_0$

At point C: $(0.125 I_0) (\cos 30^\circ)^2 = 0.0938 I_0$

(b) Now for the last filter $\phi = 90^\circ$ and $I = 0$

77.70 Light traveling in water strikes a glass plate at an angle of incidence of 59.0° , part of the beam is reflected and part is refracted. If the reflected and refracted portions makes an angle of 90° with each other, what is the (angle of) index of refraction of the glass?

Sol'n: Fig 77.70



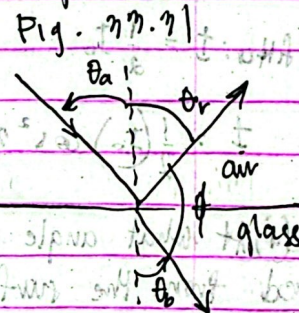
$$\theta_0 = 59^\circ, n_0 = n_a \frac{\sin \theta_a}{\sin \theta_b} = 1.33 \frac{\sin 59^\circ}{\sin 31^\circ} = 1.77$$

77.71 A parallel beam of unpolarized light in air is incident at an angle of 54.5° (with respect to the normal) on a plane glass surface. The reflected beam is linearly polarized. (a) What is the refractive index of the glass? (b) What is the refracted angle of the transmitted beam?

Sol'n: $\theta_p = 54.5^\circ$

$$(a) \tan \theta_p = \frac{n_b}{n_a} \rightarrow n_{\text{glass}} = n_{\text{air}} \tan \theta_p = 1.00 \tan 54.5^\circ = 1.40$$

$$(b) \sin \theta_b = \frac{n_a \sin \theta_a}{n_b} = \frac{1 \sin 54.5}{1.40} = 0.5815, \theta_b = 35.5^\circ$$



77.74 The refracted index of a certain glass is 1.66. For what incident angle is light reflected from the surface of this glass, is completely polarized if the glass is immersed in (a) air and (b) in water?

Sol'n: $\tan \theta_p = \frac{n_b}{n_a}, n_b = 1.66$

$$(a) n_a = 1.00, \tan \theta_p = \frac{1.66}{1.00} = 1.66, \theta_p = 58.9^\circ$$

$$(b) n_a = 1.33, \tan \theta_p = \frac{1.66}{1.33} = 1.25, \theta_p = 51.2^\circ$$

77.77 Unpolarized light of intensity 20.0 W/cm^2 is incident on two polarizing filters. The axis of the first filter is at an angle of 25.0° counterclockwise from the vertical (viewed in the direction the light is traveling), and the axis of the second filter is at 62.0° counterclockwise from the vertical. What is the intensity of the light after it has passed through the second polarizer?

Sol'n: The second filter, $\phi = 62.0^\circ - 25.0^\circ = 37.0^\circ$

At the first filter the intensity is $\frac{1}{2} I_0 = 10.0 \text{ W/m}^2$

The second filter is $I = I_0 \cos^2 \phi$, where $I_0 = 10.0 \text{ W/m}^2$ & $\phi = 37.0^\circ$

This gives $I = 6.78 \text{ W/m}^2$.

77.76. Three polarizing filters are stacked, with the polarizing axis of the second filter and the third filters at 23.0° and 62.0° , respectively, to that of first. If unpolarized light is incident on the stack, the light has intensity of 75.0 W/cm^2 after it passes through the stack. If the incident intensity is kept constant, what is the intensity of light after, it has passed through the stack if the second polarizer is removed?

Sol'n:

For unpolarized filter, $I = \frac{1}{2} I_0$

For polarized light on filter, $I = I_{\text{max}} \cos^2 \phi$

$$I = \frac{1}{2} I_0 (\cos 23.0^\circ)^2 (\cos [62 - 23])^\circ = 0.256 I_0 \quad I_0 = \frac{1}{0.256} = \frac{75 \text{ W/cm}^2}{0.256} = 297 \text{ W/cm}^2$$

With only the first & third polarizers, $I = \frac{1}{2} (I_0) (\cos 62^\circ)^2 = 0.110 I_0$

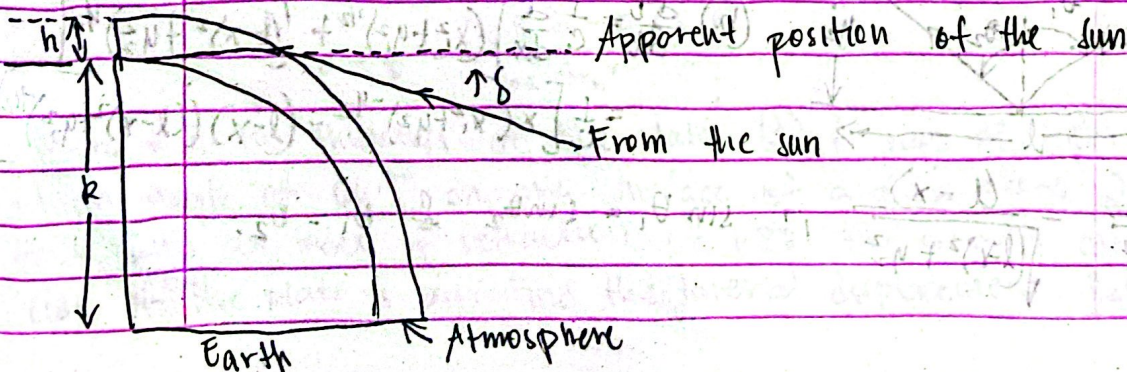
$$I = (0.110) 297 \text{ W/cm}^2 = 32.7 \text{ W/cm}^2$$

77.55 When the sun is either rising or setting and appears to be just the on the horizon, it is in fact below the horizon. The explanation for this seeming paradox, is that light from the sun bends slightly when entering the Earth's atmosphere, as shown in Fig. 77.55. Since our perception is based on the idea that light travels in straight lines, we perceive the light to be coming from an apparent position that is, an angle δ above the sun's true position. (a) Make the simplifying assumptions that the atmosphere has uniform density, and hence uniform index of refraction n , and extends to a height h above the earth's surface, at which point it abruptly stops. Show that the angle δ is given by

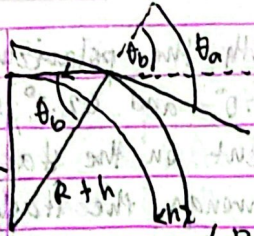
$$\delta = \arcsin\left(\frac{nR}{R+h}\right) - \arcsin\left(\frac{R}{R+h}\right)$$

Where $R = 6378 \text{ km}$ is the radius of the earth. (b) Calculate δ using $n = 1.0003$ and $h = 20 \text{ km}$. How does this compare to the angular radius of the sun, which is about one quarter of a degree?

Fig. 77.55



Sol'n: (a)



$$\delta = \theta_a - \theta_b$$

From the diagram $\sin \theta_b = \frac{R}{R+h}$

$$\theta_b = \arcsin\left(\frac{R}{R+h}\right)$$

$$\sin \theta_a = n \sin \theta_b = n \left(\frac{R}{R+h}\right) \text{ so } \theta_a = \arcsin\left(\frac{nR}{R+h}\right)$$

$$\delta = \theta_a - \theta_b = \arcsin\left(\frac{nR}{R+h}\right) - \arcsin\left(\frac{R}{R+h}\right)$$

$$(b) \frac{R}{R+h} = \frac{6.78 \times 10^6 \text{ m}}{6.78 \times 10^6 \text{ m} + 20 \times 10^3 \text{ m}} = 0.99688$$

$$\frac{nR}{R+h} = 1.0003(0.99688) = 0.99718$$

$$\delta = 0.23^\circ$$

$$\theta_b = \arcsin\left(\frac{R}{R+h}\right) = 85.47^\circ$$

$$\delta = \theta_a - \theta_b = 85.70 - 85.47 = 0.23^\circ$$

$$\theta_a = \arcsin\left(\frac{nR}{R+h}\right) = 85.70^\circ$$

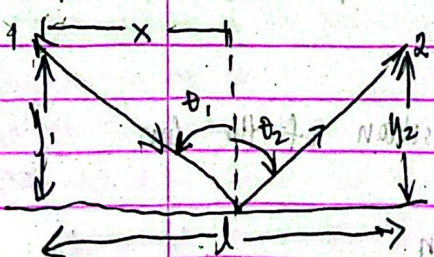
Fig. P39.56 A ray of light traveling with speed c leaves point 1, as shown in Fig. P39.56 and is reflected to point 2. The ray strikes the reflecting surface at a horizontal distance x from point 1. (a) Show that the time t required for the light travel from 1 to 2 is

$$t = \frac{\sqrt{y_1^2 + x^2}}{c} + \frac{\sqrt{y_2^2 + (l-x)^2}}{c}$$

(b) Take the derivative of t with respect to x . Set the derivative equal to zero to show that this time reaches its minimum value when $\theta_1 = \theta_2$, which is the law of reflection and corresponds to the actual path taken by the light. This is an example of Fermat's principle of least time, which states that among all possible paths between two points the one actually taken by a ray of light is that for which the time of travel is a minimum.

Figure P39.56 Sol'n: (a) $d = \sqrt{x^2 + y_1^2} + \sqrt{(l-x)^2 + y_2^2}$

$$t = \frac{d}{c} = \frac{\sqrt{x^2 + y_1^2} + \sqrt{(l-x)^2 + y_2^2}}{c}$$



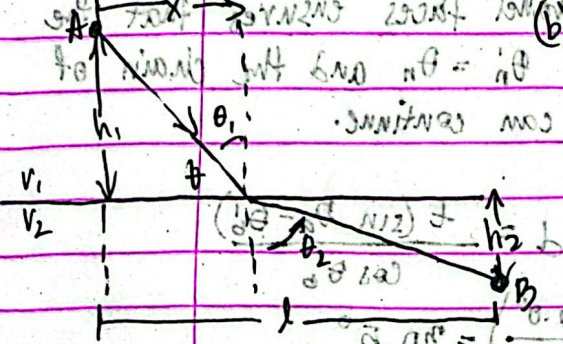
$$(b) \frac{dt}{dx} = \frac{1}{c} \frac{d}{dx} \left[\sqrt{x^2 + y_1^2} + \sqrt{(l-x)^2 + y_2^2} \right]$$

$$= \frac{1}{c} \left[x(x^2 + y_1^2)^{-1/2} - (l-x)((l-x)^2 + y_2^2)^{-1/2} \right]$$

$$\frac{x}{\sqrt{x^2 + y_1^2}} = \frac{(l-x)}{\sqrt{(l-x)^2 + y_2^2}} \quad \therefore \sin \theta_1 = \sin \theta_2 \quad \& \quad \theta_1 = \theta_2$$

Q. 57 A ray of light goes from point A in a medium in which the speed of light is v_1 to point B in a medium in which the speed is v_2 . The ray strikes the interface a horizontal distance x to the right of point A. (a) Show that the time required for the light to go from A to B is

$$t = \frac{\sqrt{h_1^2 + x^2}}{v_1} + \frac{\sqrt{h_2^2 + (l-x)^2}}{v_2}$$



(b) Take the derivative of t with respect to x . Set this derivative equal to zero to show that this time reaches its minimum value when $n_1 \sin \theta_1 = n_2 \sin \theta_2$. This is Snell's law and corresponds to the actual path taken by the light. This is another example of Fermat's principle of least time. (6)

Sol'n: (a) $t_1 = \frac{\sqrt{h_1^2 + x^2}}{v_1}$, $t_2 = \frac{\sqrt{h_2^2 + (l-x)^2}}{v_2}$

$$t = t_1 + t_2 = \frac{\sqrt{h_1^2 + x^2}}{v_1} + \frac{\sqrt{h_2^2 + (l-x)^2}}{v_2}$$

$$(b) \frac{dt}{dx} = \frac{1}{v_1} \left(\frac{1}{2} \right) (h_1^2 + x^2)^{-1/2} (2x) + \frac{1}{v_2} \left(\frac{1}{2} \right) (h_2^2 + (l-x)^2)^{-1/2} 2(l-x)(-1) = 0$$

$$\frac{x}{v_1 \sqrt{h_1^2 + x^2}} = \frac{l-x}{v_2 \sqrt{h_2^2 + (l-x)^2}} \quad \text{Multiplying both sides by } e$$

$$\text{Gives } \frac{c}{v_1} \frac{x}{\sqrt{h_1^2 + x^2}} = \frac{c}{v_2} \frac{l-x}{\sqrt{h_2^2 + (l-x)^2}} \quad \frac{c}{v_1} = n_1 \quad \& \quad \frac{c}{v_2} = n_2$$

$$\sin \theta_1 = \frac{x}{\sqrt{h_1^2 + x^2}} \quad \text{and} \quad \sin \theta_2 = \frac{l-x}{\sqrt{h_2^2 + (l-x)^2}}$$

Q. 58 Light is incident in air at an angle θ_a (Fig. P. 58) on the upper surface of a transparent plate, the surface of the plate being plane and parallel to each other. (a) Prove that $\theta_a = \theta_a'$ (b) show that is true for any number of different parallel plates. (c) Prove that the lateral displacement d of the emergent beam is given by the relationship

$$d = t \frac{\sin(\theta_a - \theta_b')}{\cos \theta_b'}$$

Where t is the thickness of the plate. (d) A ray of light is incident at an angle of 66.0° on one surface of a glass plate 2.40 cm thick with an index of refraction of 1.80. The medium on either side of the plate is air. Find the lateral displacement between the